Code No: 118BA **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year II Semester Examinations, June - 2024
DATABASE SECURITY
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a)	What is a trap door?	[2]
b)	What is the need of log journal in database security?	[3]
c)	State discretionary security property.	[2]
d)	What access modes are considered in Bell-LaPadula model?	[3]
e)	What is meant by observability of a database?	[2]
f)	What is the purpose of ASES system?	[3]
g)	List the entities of SORION model.	[2]
h)	What are the security threats for a frame based system?	[3]
i)	Define active database.	[2]
j)	What is ECA model?	[3]

PART – B

(50 Marks)

2. Summarize database protection requirements. [10]

OR

3. Make a comparison of Hsiao's model and Fernandez's model for database security. [10]

4. Describe the family of policies proposed by Biba model. [10]

OR

5. Contrast flow control mechanisms at run time with flow control mechanisms at compile time. [10]

6. Illustrate the system R authorization model for secure DBMS design. [10]

OR

7. Explain the evaluation criteria for inference controls for a statistical database protection. [10]

8. Discuss the special features of a frame based system and compare with object oriented system. [10]

OR

9.a) Contrast content constraints with context constraints.
b) Demonstrate composite objects in object oriented databases. [5+5]

10. Describe in detail Jagodia and Kogan's model for protecting an active database. [10]

OR

11. Demonstrate MLS rules and MLS active data model of an active database. [10]

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